

Detailed Notes: Gemini Configuration, Prompt Engineering and Parameters

1. Prompt Engineering in Gemini

Prompt Engineering is the process of designing effective instructions (prompts) to guide Gemini toward producing useful, accurate, and relevant outputs. Key Components: • Context – background information for the model. • Objective – what you want Gemini to do. • Constraints – formatting, length, style, tone, etc. • Examples – demonstrations that improve consistency. Good Prompt Characteristics: • Specific and clear. • Contains sufficient context. • Defines expected output format. • Uses constraints when required. Poor Prompts: • Ambiguous and vague. • Missing context. • No clear objective. Benefits: • Better accuracy. • Reduced hallucinations. • Consistent outputs. • Improved task performance.

2. Understanding Gemini API

Gemini API allows developers to integrate Gemini models into applications. Workflow: 1. Authentication using API Key. 2. Send request from application. 3. Gemini model processes request. 4. Response returned in JSON format. Capabilities: • Text Generation • Code Generation • Reasoning • Vision/Image Understanding • Data Analysis Typical Applications: • Chatbots • Virtual Assistants • Content Generation Tools • Analytics Platforms • Educational Systems

3. System Prompt vs User Prompt

System Prompt: Defines permanent behavior of the AI. Examples: • Assistant personality. • Response style. • Safety rules. • Global instructions. User Prompt: The specific task requested by the user. Examples: • Summarize a document. • Write a poem. • Explain robotics. Differences: System Prompt = Rules and behavior. User Prompt = Current task and context. Priority: System Prompt generally has higher priority than user instructions.

4. Gemini Parameters Overview

Three major parameters influence output generation: Temperature: Controls randomness and creativity. Top-K: Limits candidate token selection to the K most probable tokens. Top-P (Nucleus Sampling): Selects tokens from the smallest probability set whose cumulative probability reaches P. Together these parameters balance: • Creativity • Accuracy • Diversity • Predictability

5. Temperature Parameter

Temperature determines how deterministic or creative the model becomes. Low Temperature (0.0 – 0.3): • Predictable • Logical • Consistent • Suitable for coding, factual tasks and summaries Medium Temperature (0.4 – 0.7): • Balanced creativity and accuracy • General-purpose usage High Temperature (0.8 – 1.5): • More creative • Diverse wording • Less predictable • Useful for storytelling and brainstorming Examples: Low temperature generates focused responses. High temperature generates imaginative responses.

6. Low vs High Temperature Outputs

Low Temperature Output: • Repeatable results. • Similar answer every time. • High confidence and precision. • Useful in technical applications. High Temperature Output: • Varied responses. • Rich language. • Creative exploration. • Better for content creation. Comparison: Low Temp → Accuracy, Logic, Consistency. High Temp → Creativity, Diversity, Surprise.

7. Understanding Top-K Sampling

Top-K sampling restricts selection to the K most probable tokens. Process: 1. Model predicts probabilities for all possible next tokens. 2. Tokens are sorted by probability. 3. Only top K candidates remain. 4. Random selection occurs among those K tokens. Example: K = 5 Only five highest probability words can be selected. Effects: Small K: • Focused output • More deterministic Large K: • More creativity • Greater diversity Advantages: • Prevents extremely unlikely token selection. • Maintains coherence.

8. Understanding Top-P Sampling

Top-P (Nucleus Sampling) dynamically selects tokens. Process: 1. Sort tokens by probability. 2. Add probabilities until cumulative probability reaches P. 3. Only tokens inside this nucleus are retained. 4. Randomly choose among them. Example: P = 0.90 Model selects from tokens covering 90% cumulative probability. Advantages: • Dynamic candidate pool. • Better balance between creativity and coherence. • Adapts to probability distribution. Smaller P: • Safer and more focused. Larger P: • More diverse and creative.

9. Relationship Between Temperature, Top-K and Top-P

Temperature controls randomness. Top-K controls candidate count. Top-P controls cumulative probability coverage. Recommended Settings: Coding: Temperature = 0.1–0.3 Top-K = Small Top-P = 0.8–0.9 Chatbots: Temperature = 0.5–0.7 Top-K = Medium Top-P = 0.9 Creative Writing: Temperature = 0.9–1.2 Top-K = Large Top-P = 0.95–1.0

10. Practical Example from Slides

Question: Who is the next best Avenger? If Top-K = 2: Only the two highest probability candidates are considered. If Top-P = 0.80: The model includes enough candidates to cover 80% total probability. Low Temperature: Likely chooses highest probability answer repeatedly. High Temperature: May choose alternative candidates for diversity. Conclusion: The final output depends on the combined interaction of Temperature, Top-K and Top-P.